Alphabet (GOOGL / GOOG) | In-depth research report

Research date: 13 November 2026, Hong Kong time. Historical financials cover fiscal 2016-2025. Operating results run through Q3 2026, with company and customer disclosures checked through 13 November. The valuation uses the 9 November frozen model. The market comparison uses the 6 November regular U.S. close, $284.78 for GOOGL and $283.93 for GOOG, with share counts as of 21 October. This research example brings together AI Berkshire and AIVA's long-term study, four specialist updates, two contested assumptions and their cash-flow sensitivities.

Alphabet's three main businesses have different economics. Search & other makes up roughly half of group revenue. Its principal offering to merchants is measurable access to prospective customers. YouTube shares revenue with creators to support the supply of content, then charges advertisers and subscribers. Google Cloud earns recurring fees for computing and software and also sells TPU systems that can be delivered in batches.

Alphabet still attracts users and enterprise demand. The main valuation question is how much capital it takes to serve that demand. Operating income compounded at 20.7% from 2016 to 2025, versus 12.3% for operating cash flow less cash capital expenditure. In the first nine months of 2026 those two cash figures were $127.149 billion and $131.953 billion. Anthropic's minimum hosting spend makes some demand easier to see. Delivery, collections, credit support and equity investments determine what Alphabet ultimately keeps. The evidence supports keeping both frozen model inputs for the sensitivity tests. Alphabet 10-Q · Anthropic S-1/A

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1. Company, ownership and development

Alphabet is the listed U.S. parent based in Mountain View, California. Google is its largest operating business. Listed Class A shares (GOOGL) carry one vote each, unlisted Class B shares carry ten, and listed Class C shares (GOOG) generally carry none. An equity valuation must cover all three classes once. Sundar Pichai leads Alphabet and Google. The 2026 proxy lists John Hennessy as chair and Anat Ashkenazi as CFO. At the proxy record date, 6 April 2026, Larry Page and Sergey Brin held 27.4% and 25.3% of voting power respectively, or 52.7% together. BlackRock's cited 6.1% relates to Class A ownership and represents 2.5% of voting power. The underlying BlackRock filing was dated 12 February 2024. Alphabet overview · 2026 proxy

Google Services includes Search, YouTube, subscriptions, platforms and devices. Google Cloud includes enterprise infrastructure, Workspace and related products. Other Bets includes businesses at different stages, such as Waymo. Alphabet also reports shared AI research and other activity at group level. This matters when moving from segment operating profit to consolidated cash. 2025 10-K

How the business developed

Date	Event	Why it matters
1996-1998	A Stanford research project became Google	Search became the recurring consumer entry point
August 2004	Google listed its shares	Access to public capital alongside founder voting control
2005	Google acquired Android	Mobile operating systems and app distribution brought Google products into more daily activities
November 2006	Google completed its YouTube acquisition	Video added a creator, audience and advertising ecosystem
March 2008	Google completed the DoubleClick acquisition	The company entered advertising infrastructure beyond its own search pages
2014	Google announced its DeepMind acquisition	A long-term foundation for model research
2015	The Alphabet reorganization was announced	Google and the exploration businesses gained clearer operating boundaries
2016	Google disclosed TPU, after more than a year of internal use	The company began to show how custom computing infrastructure could support machine learning
December 2019	The founders left their executive posts and Pichai became CEO of both companies	Operating responsibility concentrated while the founders retained voting control
April 2023	Google Brain and DeepMind combined	Core model development came under one organization
March 2026	Alphabet completed the Wiz acquisition	Cross-cloud security extended the enterprise relationship beyond compute
First nine months of 2026	Alphabet issued common stock, preferred stock and debt, and did not repurchase common shares	Growth funding and the common share count now matter more to per-share returns

Google history · Alphabet reorganization · 2019 leadership transition · TPU announcement · AI team reorganization · Wiz completion

Financial scale

Amounts in USD billions. The full year, nine-month cumulative period and trailing twelve months use different measurement windows.

Metric	FY 2025	First nine months of 2026	TTM to Q3 2026
Revenue	402.836	357.269	471.098
Operating income	129.039	121.042	156.976
Net income	132.170	201.239	235.694
Operating cash flow	164.713	127.149	179.551
Cash capex	91.447	131.953	159.804
Operating cash flow less cash capex	73.266	-4.804	19.747

Trailing twelve months equal fiscal 2025 plus the first nine months of 2026 less the comparable 2025 period. The cash residual is a simple free-cash-flow measure before economic adjustments for stock compensation, leases and financing. Q3 revenue was $127.577 billion and operating income was $40.576 billion. Investment gains boosted trailing net income well above operating income. Capitalizing reported net income would therefore risk counting investment gains and then adding the same assets again.

At 30 September 2026, Alphabet held $56.334 billion of cash and equivalents and $265.098 billion of cash and marketable securities, including equities. Borrowings totaled $126.118 billion before finance leases. For the first nine months, net proceeds from common and mandatory convertible preferred stock were $42.870 billion and $19.063 billion. Debt issuance brought in $85.222 billion, and debt repayments used $8.879 billion. A larger cash balance can reflect financing as well as operating cash generation. 2025 10-K · 2026 Q3 10-Q

2. Business model and unit economics

Consumers and enterprises return to Alphabet products to complete recurring tasks. Advertisers pay to reach those users. Consumers pay for selected subscriptions. Enterprises pay for computing, software and other services. The value drivers are the high-value tasks Alphabet retains, the revenue each task supports, and the operating and capital cost of delivering it.

Revenue mix

USD billions for Q3 2026. The denominator is consolidated revenue of $127.577 billion.

Revenue source	Revenue	Share of total
Search & other	63.350	49.7%
YouTube ads	11.207	8.8%
Google Network	7.455	5.8%
Subscriptions, platforms and devices	14.290	11.2%
Google Cloud	30.709	24.1%
Other Bets	0.372	0.3%
Revenue hedging gains	0.194	0.2%
Total	127.577	100%

The three advertising lines total $82.012 billion, or 64.3% of consolidated revenue. The first four lines make up Google Services, roughly 75.5%. Subscriptions, platforms and devices include products beyond YouTube. Alphabet 10-Q, Note 2

Search: the advertiser pays for a chance to win business

A search user generally pays nothing per query. An advertiser pays for an impression, click or other eligible interaction in the hope of obtaining a sale, lead or awareness. The economic variables are the number of monetizable interactions and their price. Both depend on intent, advertiser budgets, ad quality and competition. The auction also considers landing-page quality, thresholds, context and expected impact of ad assets.

As a teaching example, 100 clicks cost 100 currency units. If the clicks produce five incremental orders, each contributing 30 after product and fulfillment costs, the advertiser keeps 50 after advertising. Two orders at the same economics leave a loss of 40. Better matching and measurable incremental conversion can support repeat spending. Google also pays to acquire traffic through partners and spends on serving, developing and distributing the product. Ad Rank · Conversion value and ROAS

AI Search: measure the whole task

Someone shopping for a dishwasher may start by checking dimensions and finish by comparing brands, installation and prices. AI Mode can hold those steps in one conversation, giving Google a chance to identify buying intent earlier and serve an ad closer to the decision. The same feature can also replace several monetizable searches with one answer and raise the cost of serving the task.

Google announced Gemini 4 Carbon for eligible AI Ultra subscribers in Search AI Mode on 10 November. It announced a global rollout of Gemini 4 Hydrogen in Search on 12 November, with more complex intent handling, parallel searches and structured comparisons. Google Ads reporting aggregates those placements into existing campaign and search-term reporting. The product can reuse advertiser demand, while researchers have limited visibility into placement-level incremental conversion. Carbon announcement · Hydrogen announcement · Google Ads Help

For existing tasks that move to AI, multiply the number of migrated tasks by the change in revenue per task. For genuinely new tasks, multiply their number by AI revenue per task. Deduct incremental operating costs and account for required reinvestment in the cash-flow bridge. Management said most AI Mode queries were incremental and also acknowledged lower monetization in AI Mode. More follow-up queries can arise within a single task. Comparable task-level conversions, merchant economics and Alphabet's serving costs

would allow a better test. Q3 earnings call

YouTube: creator economics support the supply of content

Creators provide content, audiences attract advertisers, and payments support further production. Under the applicable partner modules, creators receive 55% of eligible watch-page net ad revenue, 45% of their allocation from the Shorts creator pool and 70% of eligible net fan-funding revenue. Those percentages have different denominators. Video delivery, storage, moderation, recommendations and rights costs also matter. Premium brings subscription revenue and shares it with creators, while YouTube TV and sports can carry separate content obligations. YouTube partner earnings · YouTube Premium revenue

Cloud: a recurring purchasing relationship and a hardware business

GCP sells infrastructure, data, AI and security, often by usage or subscription. Workspace is mainly subscription software embedded in work processes. Moving data, permissions and workloads creates real switching costs. Open tools and multi-cloud deployment give customers more choice for incremental work. Wiz expands security coverage across clouds, with economic returns depending on cash earned after the acquisition cost.

Hosted compute requires capacity, power and networking before revenue. TPU system sales can shift some facility costs to customers, while Alphabet still finances inventory through delivery and collection. Management expected temporary third-party capacity to put pressure on near-term Cloud margins as depreciation and operations costs also grew. The 2026 capital expenditure guidance was $195-$205 billion. Management indicated a significant increase for 2027 without specifying a number. Alphabet 10-Q · Q3 earnings call · Wiz announcement

Delivery model	Alphabet revenue	Capital and cash obligations
Google Cloud hosting	Usage, contracted capacity or service delivered over time	Equipment, facilities or leases precede cash recovery. Utilization, pricing and renewals drive returns
Direct system sale	Hardware and associated software on transfer of control, then service as delivered	Customer may supply the facility. Alphabet still carries inventory, collection timing and sometimes financing or repurchase obligations
Deployment through another operator	Equipment, service or channel revenue under the agreement	Power, cooling, network and equipment responsibility depend on the contract

Google announced TPU 8tx and 8i Max on 9 November and said systems can be deployed in customer data centers. Oracle earlier described a proposed deployment in which OCI provides power, cooling and networking. Alphabet disclosed maximum potential payments of $6.6 billion under certain TPU repurchase guarantees at 30 September, with the obligations' fair value immaterial. Actual loss depends on triggering conditions and recovered equipment value. Google Cloud TPU announcement · Oracle announcement · Alphabet 10-Q, Note 10

3. Ten years of financial statements and cash flow

The ten fiscal-year observations run from 2016 to 2025, with nine intervals for compound growth. USD billions except where indicated. Annual filings in the appendix provide the original source for each year.

Revenue, profit and cash flows

Fiscal year	Revenue	Operating income	Net income	Operating cash flow	Cash capex	Simple FCF
2016	90.27	23.72	19.48	36.04	10.21	25.82
2017	110.86	26.15	12.66	37.09	13.18	23.91
2018	136.82	26.32	30.74	47.97	25.14	22.83
2019	161.86	34.23	34.34	54.52	23.55	30.97
2020	182.53	41.22	40.27	65.12	22.28	42.84
2021	257.64	78.71	76.03	91.65	24.64	67.01
2022	282.84	74.84	59.97	91.50	31.48	60.01
2023	307.39	84.29	73.80	101.75	32.25	69.50
2024	350.02	112.39	100.12	125.30	52.53	72.76
2025	402.84	129.04	132.17	164.71	91.45	73.27

Annual figures use each year's original 10-K. The simple cash measure equals operating cash flow less cash purchases of property and equipment. A full owner-cash measure also needs to account for stock compensation, leases and noncommon claims. A 2019 reporting change moved investment performance fees out of operating costs, increasing restated 2017 operating income from $26.146 billion to $26.178 billion and 2018 from $26.321 billion to $27.524 billion, without changing total net income or cash. On a consistent presentation, 2018-2019 operating income growth was about 24.4% rather than the 30.1% calculated from original figures. 2019 10-K

Capital intensity, share count and cash per share

Fiscal year	Operating margin	Cash capex / sales	Stock compensation	Year-end shares, split-adjusted (bn)	Simple FCF per share (USD)
2016	26.3%	11.3%	6.70	13.826	1.87
2017	23.6%	11.9%	7.68	13.896	1.72
2018	19.2%	18.4%	9.35	13.911	1.64
2019	21.1%	14.5%	10.79	13.767	2.25
2020	22.6%	12.2%	12.99	13.504	3.17
2021	30.6%	9.6%	15.38	13.242	5.06
2022	26.5%	11.1%	19.36	12.849	4.67
2023	27.4%	10.5%	22.46	12.460	5.58
2024	32.1%	15.0%	22.79	12.211	5.96
2025	32.0%	22.7%	24.95	12.088	6.06

The share counts reflect the split on a consistent basis. Simple cash per share uses year-end shares and differs from diluted EPS. Share count declined over much of the period, while stock compensation and capital spending rose.

Balance sheet

Fiscal year-end	Cash and equivalents	Cash, equivalents and short-term investments	Total assets	Shareholders' equity
2016	12.92	86.33	167.50	139.04
2017	10.71	101.87	197.29	152.50
2018	16.70	109.14	232.79	177.63
2019	18.50	119.67	275.91	201.44
2020	26.46	136.69	319.62	222.54
2021	20.95	139.65	359.27	251.63
2022	21.88	113.76	365.26	256.14
2023	24.05	110.92	402.39	283.38
2024	23.47	95.66	450.26	325.08
2025	30.71	126.84	595.28	415.26

2016-2020. Revenue roughly doubled from $90.3 billion to $182.5 billion and operating cash flow rose from $36.0 billion to $65.1 billion. The $12.7 billion net income in 2017 includes about $9.9 billion of net tax expense from tax reform. Investment fair-value movements also affected later net profit. 2017 10-K

2021-2023. The 2021 revenue increase was 41.2%, and the operating margin rose to 30.6% before falling to 26.5% in 2022. Extending server and network asset lives reduced depreciation by $2.6 billion in 2021 and $3.9 billion in 2023, increasing reported net income by about $2.0 billion and $3.0 billion respectively. These changes affected accounting expense without increasing current cash receipts. DeepMind moved from Other Bets to Alphabet-level activities in 2023. Comparisons of segment operating income require a consistent disclosure version. 2021 10-K · 2023 10-K

2024-2025. In 2025, revenue rose 15.1%, operating income 14.8%, and operating cash flow 31.5%. Cash capex rose 74.1% and reached 22.7% of revenue. Of the $39.414 billion increase in operating cash flow, $38.912 billion went to higher cash capex, leaving a $0.502 billion improvement in simple free cash flow. The operating-income compound growth rate over the full period was 20.7%, versus 12.3% for simple free cash flow. 2025 10-K

Latest operating results and the investment cycle

In Q3 2026, Services generated $96.302 billion of revenue and $39.157 billion of segment operating income. Cloud generated $30.709 billion of revenue and $11.121 billion of segment operating income. Other Bets generated $0.372 billion of revenue and an operating loss of $2.047 billion. Alphabet-level activities and unallocated costs absorbed $7.655 billion. Consolidated operating income was $40.576 billion.

Search & other revenue rose about 12%, alongside 8% higher paid clicks and 4% higher cost per click for the whole category. Management cited currency as the largest factor in deceleration relative to Q2, along

with sector-specific effects and lower AI Mode monetization. Cloud revenue increased approximately 103% from $15.157 billion. The company identified core GCP, AI and TPU systems as drivers, without quantifying the hardware contribution separately. Temporary third-party capacity was expected to constrain near-term margins.

Full-year 2026 capex guidance of $195-$205 billion, less the first nine months' $131.953 billion, implies around $63-$73 billion in Q4. The first nine months' simple cash residual was negative $4.804 billion. Assets yet to be placed in service rose from $78.592 billion at year-end 2025 to $145.270 billion at September 2026. That investment-to-operation lag is observable. Future utilization, margins and equipment replacement will determine the cash outcome. At September end the group also disclosed $924.4 billion of purchase commitments, including $213.4 billion due over the following twelve months, and $130.5 billion in payments under leases not yet commenced. These are multi-period commitments across several cost categories and require care to avoid double counting with the cash forecasts. Alphabet 10-Q, Notes 7 and 15 and Future Cash Commitments · Q3 earnings call

4. Moat, competition, culture and management

Search combines habitual use, distribution, information quality and an advertising marketplace. Changing a website is easy. Replicating all four at scale and delivering measurable merchant returns is harder. Paid distribution also has a cost. YouTube's moat depends on creators and audiences returning to one another, and on viable revenue sharing. Cloud's moat depends on data, software, identity and deployment integration. Customers can still choose other platforms for incremental workloads.

Business	Direction since roughly 2021	Evidence that could change the view
Search	Broader product capability, more competition for the default task	Commercial queries and ad budgets move away, or higher monetization fails to pay for serving and distribution
YouTube	Meaningful creator-audience feedback loop	High-value creators or audiences leave, or preserving the supply requires persistently higher sharing and rights costs
Cloud	More embedded in data, permissions and workflows, with heavier capital commitments	Replacement needs, inventories, receivables and price pressure grow faster than the cash earned

Management's record combines patient engineering and a willingness to restructure. The founders' IPO letter stressed long-term projects. The internally developed TPU project, the Alphabet reorganization and the AI team combination show how that approach was put into practice. Stadia's closure and refund plan show the company can end an unsuccessful product. Founder voting control also limits other shareholders' ability to change the capital allocation strategy. Long-term executive awards partly link management to shareholder returns, including unit-value provisions for Waymo and Wing. The investment test remains what the decisions deliver per common share. Founders' IPO letter · Stadia announcement · 2026 proxy

Google's code of conduct addresses fair competition. The court found that exclusive distribution arrangements helped maintain the search monopoly. Distribution and legal remedies affect access and its cost, alongside continuing product strengths. Company policy and judicial findings provide different kinds of evidence about conduct. Google code of conduct · U.S. court opinion

AI spending reaches chipmakers, power and network suppliers, cloud operators, model providers and customer-facing applications. Alphabet participates in several layers. Integration can improve utilization and keep more of the margin, while the same structure exposes Alphabet to research, replacement and idle capacity. AWS reported $45.984 billion of revenue and $17.773 billion of operating income in the same quarter, with about 39.3% revenue growth. Segment figures indicate demand and competitive scale. They have different asset and cost boundaries. Amazon Q3 results

Anthropic: demand, equity investment and credit exposure

Anthropic committed approximately $121.3 billion in minimum Google hosting spend from April 2026 to July 2033. The agreement provides for a true-up if spend falls short, subject to adjustments and exceptions, with cancellation available in the event of Google's uncured material breach. The 26 October initial filing reported about $114.5 billion. The 2 November amendment showed $121.3 billion, and the 10 November amendment confirmed that amount. Separate TPU procurement may use Google hosting, direct purchases with Google financing, or third-party purchases or leases. Anthropic can allocate incremental workloads across architectures.

The 10 November amendment disclosed a signed agreement for Google's proposed $10 billion concurrent IPO private placement, subject to closing conditions. It uses half the previously available $20 billion under the arrangement. TeraWulf separately disclosed conditional Google credit support for specified Anthropic data-center leases. Follow service delivery, invoicing and receipts in the operating cash record. Follow the subscription payment and investment returns separately. Assess each guarantee on its own triggers and recoveries. The hosting commitment cannot be reconciled contract by contract with Alphabet's disclosed remaining performance obligations. Adding the totals could double count revenue. 26 October S-1 · 2 November S-1/A · 10 November S-1/A · TeraWulf 10-Q, Note 12

The strongest bear case accepts that Alphabet can retain users and grow reported earnings. The concern is that maintaining those gains may absorb enough computing and replacement capital to delay common-share cash returns. The strongest rebuttal is the lag between paying for capacity and earning cash on it. If commissioned assets earn attractive margins, collect as contracted and can be maintained at reasonable cost, current spending can support future returns.

5. Valuation, sensitivity and the bull/bear exchange

5.1 Valuation object and method

The analysis values Alphabet's existing common equity. Start from operating cash available to all capital providers, then add nonoperating assets and deduct debt, leases, preferred claims, minority interests and other claims. Trailing operating income was $156.976 billion versus $235.694 billion of net income. The operating forecast excludes investment gains and the equity bridge includes the relevant investments. FCFF avoids assuming a detailed future borrowing and preferred-conversion schedule just to value the operating business.

The 9 November rebuild tested 35 cases across three public implementations. The operating forecast is project code, with external implementations used to check formulas.

Candidate	Test result	Use in this case
valuation_course	The stable-growth reinvestment identity worked. The project lacked a complete operating forecast and full input validation.	Cross-check the formula.
valuation_analyst	The standard route deducted reinvestment from already-net cash. The coherent route can use negative-to-positive cash flow and rebuild terminal cash.	Check specified cash flows through the coherent route.
dcf-valuation-tool	Fixed ratios did not fit the build phase. The transition did not reset working-capital growth.	Use the test to illustrate normalization.

A simplified company has last-year revenue of 1,100, NOPAT of 165 and added working capital of 20, leaving cash of 145. If growth falls to 2%, next-year NOPAT reaches 168.3 and added working capital falls to 4.4, leaving cash of 163.9. Multiplying the last cash flow by 1.02 would give 147.9. Growth and reinvestment · Terminal value and excess returns

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NOPAT = EBIT × (1 - operating cash tax rate)
Net reinvestment = cash capex + new finance leases - depreciation
                 + increase in operating working capital
FCFF = NOPAT - net reinvestment
Stable reinvestment rate = stable growth / incremental ROIC
Terminal FCFF = next-period NOPAT × (1 - stable growth / incremental ROIC)
Terminal value = terminal FCFF / (WACC - stable growth)
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For example, a company with 100 of after-tax operating profit and 3% growth needs 20 of reinvestment at a 15% incremental return, leaving 80 of cash. At a 10% incremental return it needs 30, leaving 70. Alphabet's historical return on its Search franchise does not determine the return on the next server fleet or hardware inventory.

5.2 Operating forecast and capital phases

Services and Cloud revenues are projected separately, with shared AI and corporate costs and Other Bets losses retained at the group level. Operating margins include research and future stock compensation. Consistent 2023-2025 segment revenues were $272.543, $304.930 and $342.721 billion for Services and $33.088, $43.229 and $58.705 billion for Cloud.

During the build, cash capex, leases, depreciation and working capital are explicit. The forward years are rolling twelve-month periods, not management's calendar-year guidance. During capital recovery, divide additional annual sales by the marginal sales-to-capital ratio (S/C). An S/C of 0.8 uses $1.25 of net operating capital per extra dollar of annual revenue, while 1.2 uses about $0.83. Company-wide historical proxies for 2023-2025 were 0.97, 0.98 and 0.78. Excluding a broad accrued-liability movement reduces the 2025 proxy to about 0.65. Timing, business mix and broad working-capital measures limit comparability.

From Y4 through Y7, the model blends the declining construction and working-capital path with normal reinvestment. Construction weights are 80%, 60%, 40% and 20%, then zero in Y8. The tail uses Y3 net fixed investment decaying 5% a year, plus working capital. The normal path uses incremental sales divided by S/C. The weights combine alternative estimates of total net investment. From Y9 the ratio converges toward one implied by stable incremental ROIC and operating margin. Alphabet filings

5.3 Asset commissioning, depreciation and replacement

At 30 September 2026, the original cost of in-service assets was $341.100 billion and assets not yet in service were $145.270 billion. Deducting $113.543 billion of accumulated depreciation gives $372.827 billion net PP&E. The commissioning model uses the pending asset pool to study when depreciation begins and the potential cost of replacing assets.

Input	Source or assumption
Pending asset pool of 145.270	Q3 10-Q Note 7, in USD billions
60% servers and networking	Proxy drawn from disclosed in-service technical infrastructure. The pending pool mix is unknown
Four- or six-year server life	Six years reflects the general book policy. Four years tests faster replacement
20- or 30-year long-lived assets	Model assumptions. Some land is not depreciated
Faster 50%/35%/15%; slower 20%/40%/40%	Three-year commissioning scenarios
Midyear commissioning and half-year depreciation	Model convention

Commissioning case	Year 1	Year 2	Year 3
Faster commissioning	4.02-6.17	10.85-16.67	14.87-22.84
Slower commissioning	1.61-2.47	6.43-9.88	12.86-19.76

Values are USD billions of additional depreciation from the original pending pool. The illustration treats the September pool as a starting 2027 pool and excludes existing in-service assets, new procurement and intangibles. Same-price long-run replacement of this modeled pool is approximately $16.08-$24.70 billion per year. Timing, residual values, utilization and prices can change actual cash needs. Accelerating accounting depreciation at fixed pre-depreciation earnings lowers EBIT. Accelerating economic retirement also raises cash replacement. The cash-to-asset bridge has a negative $14.483 billion residual after comparison with purchases, financing leases and changes in unpaid equipment purchases. It can include disposals, acquisitions, currency, prepayments and scope differences. Alphabet 10-Q, Note 7

5.4 What the base assumptions require

Driver	Historical anchor	Base path	Evidence for revision
Services growth	12.39% in 2025, 13.62% YTD, 10.62% in Q3	12%, 10%, 9%, then converges to 3%	Sustained shifts in commercial queries, conversion or advertiser returns
Cloud growth	35.80% in 2025, 83.97% YTD, 102.61% in Q3	40%, 30%, 25%, 20%, then converges to 3%	Mix of hardware and recurring consumption, collections and customer financing
Segment margins	TTM Services 42.36%, Cloud 34.18%	Services 41.5% to 38%; Cloud 32%, briefly 33%, ultimately 30%	AI unit cost, ad revenue per task, equipment margins and utilization
Shared costs and Other Bets	About 5.25% and 2.03% of trailing revenue	Shared 5.5% to 4%; Other Bets loss 1.8% to 0.5%	Spending grows with sales without corresponding operating returns
Cash capex and leases	2026 company capex guidance 195-205 in USD billions; 2027 significant increase without a figure	250/280/270 of cash capex plus 5/6/7 new leases over Y1-Y3	Spending raises, project delays or idle capacity
Working capital	Broad nine-month cash-use proxy 35.132 in USD billions	15% of additional Services revenue and 65% of additional Cloud revenue	Inventory, receivables and customer prepayments
Mid-period S/C	0.97/0.98/0.78 proxies in 2023-2025	Transitions to 0.9 as capacity starts earning revenue	Repeated additional investment needed for the same growth
Stable return	No observable perpetual marginal return	3% nominal growth, 12% incremental ROIC, 10% WACC	Incremental returns converge to or fall below funding costs

The first three years' depreciation uses a $27.995 billion run-rate for existing assets, the 50%/35%/15% pending pool schedule and 50%/50% commissioning of new purchases in the current and next year. The short-life proxy uses six years and the long-life proxy thirty.

Period	Existing asset run-rate	New depreciation from pending pool	Depreciation from future purchases	Total PP&E depreciation	Implied pre-depreciation operating margin
Y1	27.995	4.116	7.225	39.336	38.94%
Y2	27.995	11.113	29.778	68.886	42.22%
Y3	27.995	15.229	61.058	104.282	46.32%

The model requires pre-depreciation operating margin of about 46.32% by Y3 versus 39.26% trailing. It therefore requires utilization, cost absorption and monetization to improve together. Holding pre-depreciation margins at the trailing level is one of the stress cases.

5.5 From revenue to year-one cash

Y1 Services revenue equals 376.341 × 1.12 = 421.502. Cloud equals 93.169 × 1.40 = 130.437. Applying segment operating margins of 41.5% and 32% yields 174.923 and 41.740. Group revenue including Other Bets is 553.627. Subtracting 30.449 shared costs and 9.965 of Other Bets losses gives 176.248 group EBIT. A 20% operating tax assumption yields 140.999 NOPAT, all in USD billions.

Y1 net reinvestment = 250 cash capex + 5 new finance leases
- 39.336 depreciation + 30.998 working capital
= 246.662
Y1 FCFF = 140.999 - 246.662 = -105.664

The table shows the full operating and cash path. The final column is realized incremental sales divided by total model reinvestment, including the construction tail.

Period	Services sales	Cloud sales	Group EBIT margin	NOPAT	Net reinvestment	FCFF	Realized sales / investment
Y1	421.5	130.4	31.8%	141.0	246.7	-105.7	0.33
Y2	463.7	169.6	31.4%	159.4	248.9	-89.5	0.33
Y3	505.4	212.0	31.8%	183.2	206.5	-23.4	0.41
Y4	545.8	254.4	31.7%	203.6	176.6	27.0	0.47
Y5	584.0	297.6	31.4%	222.3	150.1	72.2	0.54
Y6	619.1	342.2	31.3%	241.3	126.1	115.1	0.63
Y7	650.0	383.3	30.8%	255.3	98.5	156.8	0.73
Y8	679.3	421.6	30.7%	271.2	75.2	196.1	0.90
Y9	706.4	455.4	30.3%	282.0	74.7	207.4	0.82
Y10	731.2	482.7	30.5%	297.2	71.1	226.1	0.73
Y11	753.1	506.8	30.2%	305.1	71.1	234.1	0.65
Y12	775.7	527.1	30.2%	315.3	75.9	239.4	0.57
Y13	799.0	542.9	30.2%	324.8	81.2	243.6	0.48
Y14	822.9	559.2	30.2%	334.5	83.6	250.9	0.48
Y15	847.6	576.0	30.2%	344.5	86.1	258.4	0.48

5.6 Terminal value, discount rate and equity bridge

Y15 revenue is 1,426.8, NOPAT is 344.540 and FCFF is 258.405, in USD billions. At 3% growth Y16 NOPAT reaches 354.876. A 12% incremental ROIC implies reinvesting 25% of NOPAT, or 88.719, leaving terminal FCFF of 266.157. At 10% WACC the Y15 terminal value is 3,802.241, or 910.226 discounted to the valuation date. Adding 644.541 of explicit-period present value yields operating value of 1,554.767. Terminal value contributes 58.5% of operating value. Normalized S/C is about 0.482. The cash flow and reinvestment transition at the terminal boundary both grow at 3%.

The 10% nominal USD WACC was calibrated using a 4.84% U.S. ten-year yield on 5 November, beta near 1.226 and a 4.41% implied equity risk premium on 1 November, giving about 10.25% equity cost. Debt cost used a one-point spread and a 20% tax assumption. Betas, debt values and preferred valuations are proxies. The matrix shows conditional USD per share when the stable return and discount rate change together.

WACC / incremental ROIC	8%	10%	12%	14%	16%
9.0%	144.67	157.17	165.63	171.75	176.41
9.5%	130.34	141.31	148.75	154.14	158.25
10.0%	118.11	127.82	134.40	139.18	142.82
10.5%	107.59	116.21	122.06	126.32	129.56
11.0%	98.43	106.13	111.36	115.17	118.08

USD billions except per share. Common share count is in billions. Cash used in operations is reserved. Net equity investments incorporate liquidity and tax haircuts. Preferred claims are deducted once, with the base share denominator excluding prospective preferred conversion.

Item	Adjustment or result
Cash and debt securities	174.714
Operating cash reserve (assumption)	-30.000
Listed equity at 75% of 90.384	67.788
Unlisted equity at 50% of 135.230	67.615
Borrowings	-126.118
Finance lease liabilities	-3.705
Noncontrolling interest proxy	-6.900
Accrued legal matters proxy	-13.800
Historical tax liabilities proxy	-12.219
Recognized credit derivative liability	-0.606
Preferred liquidation proxy	-19.250
Net equity bridge	97.519
Operating enterprise value	1,554.767
Common equity value	1,652.286
Common shares (bn)	12.294
Conditional value per share (USD)	134.40

The unlisted-equity haircut is 50%. Moving it to 25% or 75% changes the model by about $2.75 per share. A $10 billion change elsewhere in the bridge is about $0.81 per share. Future stock-based compensation remains in operating costs. The 284 million unvested awards include future service and unrecognized compensation, so a separate residual dilution test used 0.5%, 1% and 2%. The preferred liquidation proxy does not price the full conversion option or capped calls.

5.7 Joint scenarios and stresses

The following scenarios change several operating and discount assumptions together. Values are conditional. No probabilities are assigned.

Assumption or result	Downside	Base	Favorable
Services Y1 growth / mature margin	8%□34%	12%□38%	15%□42%
Cloud Y1 growth / mature margin	25%□25%	40%□30%	60%□35%
Cash capex Y1-Y3	250□290□290	250□280□270	270□320□340
Mid-period marginal S/C	0.6	0.9	1.2
WACC / g / incremental ROIC	11%□2%□8%	10%□3%□12%	9%□3.5%□16%
Y15 group revenue	879.2	1,426.8	2,194.5
Operating enterprise value	363.1	1,554.8	3,802.6
USD per share	37.47	134.40	317.24

Selected single-assumption stresses retain the remaining base inputs.

Stress	USD per share	Change vs base
Services / Cloud incremental working-capital rates 25% / 95%	129.44	-4.96
Cash build overrun of 20% in Y1-Y3, including the tail	118.66	-15.74
Construction tail ends in Y11 instead of Y8	128.89	-5.51
Operating costs +2 percentage points of sales	120.73	-13.67
Shared costs and Other Bets retain trailing revenue share	120.40	-14.00
Four-year accounting life, holding EBITDA and allowing tax shield	135.13	+0.73
Also add modeled server replacement cash, four years versus six	113.14	-21.26
First three periods cap EBITDA margin at trailing level	130.90	-3.49

A shorter accounting life at fixed pre-depreciation earnings affects the modeled tax shield. The additional replacement stress applies only to the original pending pool and first three years of server purchases, and may overlap maintenance implicit in S/C. It does not change the terminal return assumption. The construction overrun case assumes the extra spend produces no additional earnings.

5.8 What the 13 November debate changed

Capital efficiency. The bull emphasized minimum hosting commitments and shifting part of facility investment to customers. The bear asked whether the same cohorts yield collectible revenue or merely shift investment into inventory, receivables, leasing, financing or replacement. The bull accepted a same-age comparison across commissioned cohorts, matching recognized revenue, actual collections, gross contribution and net capital. The bear accepted that efficiency can improve even with higher total leases if net spending per dollar of incremental revenue falls after including replacement and removing one-time advances or inventory releases. Public information supports retaining 0.9 and testing 0.6 and 1.2.

Cloud growth. The bear asked how much contracted demand is already in the original revenue forecast and in published remaining performance obligations. The bull agreed the multi-year commitment needs to be mapped to delivery, recognition and collections. The bull asked whether demand won by competitors beyond the minimum should reduce contracted revenue too. The bear distinguished contracted revenue and incremental workloads, assuming contracted delivery and collections hold. The base path retains 40% initial growth. The tested paths start at 30% and 50%; the growth-rate shock fades to zero by Y13. Earlier compounding still leaves different revenue bases.

Single assumption	Downside	Base	Favorable
Mid-period sales per $1 net operating capital, with conditional USD/share	0.6 → $126.49	0.9 → $134.40	1.2 → $138.63
Initial Cloud growth, with conditional USD/share	30% → $119.50	40% → $134.40	50% → $156.68

In the efficiency test, income, operating margin and terminal value remain fixed. For the same $67.657 billion Y8 sales increase, net investment is $75.174 billion at 0.9 and $56.381 billion at 1.2. Y8 FCFF improves by $18.794 billion, with about $4.24 per share of total present-value improvement. In the Cloud stress, lowering first-period growth to 30% improves first-period FCFF by $4.215 billion because the assumed working-capital saving exceeds lost operating earnings. Y5 Cloud revenue falls from about $297.59 billion to $211.94 billion, reducing long-run conditional value by $14.90 per share. Use changes in inventory, receivables, prepayments and customer cohorts to revisit working capital and the growth path when better evidence becomes available.

5.9 Method cross-checks and the 6 November price

Discounted operating cash flow is the primary method. Economic profit uses NOPAT less WACC times opening invested capital. Under identical inputs it yields the same enterprise value, with a terminal capital-base adjustment of about $660.7 billion in this implementation. The agreement is an arithmetic check. Segment analysis clarifies drivers, but a separate sum-of-the-parts value needs better segment capital and shared-cost allocation. The earlier 18× Services and 25× Cloud multiples were withdrawn. Peer enterprise multiples are a market reference with differing businesses and measurement dates.

Company	TTM operating margin	TTM revenue growth	Displayed EV/EBIT	Comparison constraint
Meta	37.78%	27.52%	17.44	Different mix, investment cycle and group costs
Microsoft	47.10%	17.91%	24.57	Includes Office and Windows as well as Azure
Amazon	13.09%	15.35%	28.69	Includes retail and logistics alongside AWS
Oracle	34.27%	21.62%	20.08	Revenue mix, leverage and construction cycle differ

As a mechanism example, stable forward EV/EBIT equals (1 - tax rate) × (1 - g / incremental ROIC) / (WACC - g). With tax 20%, growth 3%, incremental return 15% and WACC 10%, the result is 9.14× next-year EBIT. This is a teaching example for mature economics. Economic profit and DCF · Enterprise multiples

At the 6 November share prices and 21 October share counts, consolidated common equity market value was about $3,496.343 billion, or $284.39 weighted per share. Holding other assumptions fixed, a Cloud-only reverse DCF reaches that price with roughly 77.04% starting growth and a fade by Y13. The entire higher path is required: Y5 Cloud revenue would be about $882.65 billion, Y15 Cloud revenue $3,562.16 billion and group revenue $4,413.01 billion. The reverse calculation adds capital as sales rise. The model found no price-matching solution by varying only mid-period S/C from 0.3 to 3.0 or only Cloud margin within the tested range of -10 to +35 percentage points. Market participants can instead assume a different combination of Search monetization, duration, cost of capital and Cloud returns.

Source of change	USD per share impact	Cumulative conditional value
Earlier conditional value	+0.00	197.71
Explicit-period operating profit / NOPAT	-35.46	162.25
Explicit-period net reinvestment	+3.78	166.03
Cash and returns after Y15	-31.63	134.40
WACC remains 10%	+0.00	134.40
Equity bridge unchanged	+0.00	134.40
Common shares unchanged	+0.00	134.40
Same inputs, different calculation library	+0.00	134.40

The difference from the earlier $197.71 estimate mostly reflects the profit trajectory and normalized long-run cash, not a mechanical increase in capex. The lower base of $134.40 and the favorable $317.24 case frame the conditions at the historical comparison price. The favorable case needs high Services monetization, sustained Cloud growth, margin gains and better reinvestment economics together. The most useful evidence is actual service and TPU mix, cash collection, utilization, unit operating costs and the economics of commissioned equipment.

6. External analyst views

These are public summaries and rating actions collected in early November. Full reports and models were unavailable. The outside views follow the independent valuation and do not set its inputs.

Institution	Published action	Read-through
Goldman Sachs, 29 October	Buy, target $385 from $435	Public summary cited Search comparisons and FX, Cloud demand and spending. Its reported 2027 capex estimate near $350 billion was an analyst forecast. Summary
Morgan Stanley, 29 October	Overweight, $375 from $400	Public summary focused on Search innovation and agent monetization and gave hardware a lower multiple. The cited 16× 2028 earnings related to market pricing rather than its target methodology. Summary
UBS, 29 October	Neutral, $332 from $380	Public summary was more cautious on Search, YouTube and AI monetization. TPU growth alone did not establish Cloud profitability or cash conversion. Summary
JPMorgan, 29 October	Overweight, $375 from $425	Public summary linked high spending and negative FCF to lower tolerance for execution misses. Some search softness could reflect FX or comparables. Summary
Citi, 29 October	Buy, $375 from $447	Public rating action available. The model and operating rationale were unavailable. Rating action

The central difference is how quickly additional AI and Cloud revenue becomes common-share cash and how much it costs to deliver. Public targets do not reveal enough about cash investment, normalized earnings or terminal assumptions to reproduce the firms' models. New Street listed a 2 November update behind a login. A Bank of America lead related to an earlier period and did not enter the comparison. New Street research directory

Appendix | Review scope and original filings

The 9 November baseline covers long-term financials, commissioning economics, cash valuation and three joint scenarios. The 13 November run used four research roles, two debaters and independent review of six sensitivity cases. The integrated report received a separate review of historical tables, key filings, the 15-period forecast and the original questions and answers. AI Search cash contribution per task and the hardware share of incremental Cloud revenue remain unquantified in public disclosures. The 6 November price is a historical reference. The public case page gives a worked comparison and links both language editions.

Twenty original annual filings

Google Inc. (CIK 1288776) filed the 2006-2014 reports. Alphabet Inc. (CIK 1652044) filed the 2015-2025 reports. Years refer to fiscal periods. The original filing index was checked against the SEC filing identifiers.

Fiscal year	Original SEC filing	Fiscal year	Original SEC filing
2006	10-K	2016	10-K
2007	10-K	2017	10-K
2008	10-K	2018	10-K
2009	10-K	2019	10-K
2010	10-K	2020	10-K
2011	10-K	2021	10-K
2012	10-K	2022	10-K
2013	10-K	2023	10-K
2014	10-K	2024	10-K
2015	10-K	2025	10-K